Javastylosis: a tool for computer-assisted chromatic and semantics based anastylosis of frescoes

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ABSTRACT

To solve some critical issues raised by anastylosis techniques, such as the deterioration of the original fragments due to continuous handling, and to compare alternative matching solutions for choosing the best one for real restoration, the pattern recognition and computer vision communities are proposing approaches that work on virtual copies of the fragments, providing new methodologies that help in the matching task. In this paper, we present Javastylosis, a tool that can support restorers for reassembling 2D images of damaged frescoes. With this computer-assisted methodology the user can apply chromatic and semantics principles. Each virtual fragments can be moved and rotated, and the same operations can be applied on groups of fragments. During the matching task, spurious elements can be discarded, and the final result is a digital solution of the 'puzzles' made by the artwork fragments, providing a table of coordinates useful to the restorers to correctly reposition the real fragments. The graphical interface, designed to be easy and intuitive, has been studied for and tested by people belonging to the autism spectrum but with very acute visual skills, so that inclusion opportunities can be given to this group of disadvantaged people.

CCS CONCEPTS

• Applied computing~Arts and humanities • Human-centered computing~Visualization~Visualization systems and tools • Social and professional topics~User characteristics~People with disabilities

KEYWORDS

Digital anastylosis, 2D images reassembling, Computer-assisted methodology, Blind reconstruction, Disadvantaged people inclusion

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1 Introduction

Anastylosis is an archaeological term that refers to a reconstruction technique applied to ruined monuments or artworks made using the original fragments of the object to be restored. The reconstruction of an artwork hit by disruptive natural events or wars is a very time consuming and complex task. The new digital technologies, becoming every day more innovative, effective and accessible, can positively impact the cultural heritage domain, offering important advantages, such as the independence from time and real objects constraints. In particular, digital anastylosis – giving the user the chance to operate on virtual copies – can preserve the original fragments, avoiding deterioration due to continuous handling, and offering the possibility of trying different matching hypotheses, resulting in different reassembling solutions to be compared and selected by the experts for the proper real reconstruction. From the classical physical approach, anastylosis can now benefit from automatic or semi-assisted methods.

In this paper we introduce Javastylosis, a tool that can help restorer in the frescoes' reassembling process. With Javastylosis the user can apply chromatic and semantics principles moving, grouping, rotating and matching the virtual fragments, and eventually discarding spurious elements, to reassemble a digital solution of the 'puzzles' made by the images of the fractured fragments. The result, an automatically generated table with coordinates of each fragment, can then be used by restorers to correctly reposition each real fragment. The graphical interface, designed for easy and intuitive handling of the fragments, has been studied for and tested by people belonging to the autism spectrum – with very acute visual skills – to give inclusion opportunities to this group of disadvantaged people.

Starting from the project DAFNE (Digital Anastylosis of Frescoes challeNgE), an international competition to search the best virtual solutions for digital frescoes restoration techniques dealing with simulated cases whose ground truth was known, a further step is blind reconstruction, when the original undamaged artwork appearance is not known. As example of a real application scenario, we used the Pompei's frescoes.

The paper is structured as follow: Section 2 presents a brief overview of the state of the art in the topic; Section 3 summarize the results of our previous project, DAFNE, an international competition to search the best virtual solutions for digital frescoes restoration techniques; Section 4 describes Javastylosis interface and functionalities; Section 5 present some details on the texture based intra fragments matching; finally, Section 6 draws the conclusions and further developments.

2 State of the art for 2D virtual anastylosis of frescoes

Mathematical methods and computer techniques have long been applied to aid the reassembling of fragmented objects, a very difficult and time-consuming task for archaeologists and restorers. Since 1998 a real application scenario is made by the Andrea Mantegna frescoes, located in the Ovetari Chapel of the Eremitani Church, in Padua (Italy), damaged by a bombing during the Second World War. In this case [1] the comparison with an old gray level image of the fresco before the damage was possible, but unfortunately less than 8% of the artwork was reconstructed. In the last decade, different approaches have been proposed. A broad taxonomy [2] is based on the main matching factors, consisting on color information [3], geometric outline characteristics [4] or both together (shape as well as sample colors along the border of the pieces) [5, 6, 7].

Other proposed methods have been based on 'colorization', trying to give a high fidelity re-colorization solution for destroyed artworks - a crucial issue in art restoration [8, 9] - while other examples refer to image 'completion', and focus on the cases of occlusion areas that produce regions to be completed [10, 11]. A peculiar case is the one of non overlapping images, in which images are extrapolated to cover the gap between them through alignment and inpainting [12]. In cases of transition regions in which chromatic and structural properties change gradually, a patch transform allows the implementation of image interpolation and morphing by combining inconsistent images (image 'melting') [13, 14]. Another approach is based on the analysis of fracture patterns to model a cracking process, so that patterns prediction will be possible to aid computer assisted anastylosis [15].

Recently, two very promising innovative solutions have been proposed, both computation intensive, but now supported by the computer technology new high performances. One [16] exploits a genetic algorithm approach, while a second one [17] applies machine learning and deep learning techniques. With reference to this, we proposed an international challenge to look for solutions that apply the most computer advanced techniques to the reconstruction of destroyed frescoes [18]. [19] propose interactive image reconstruction from incomplete, irregular and imprecise fragments, while a prototype of a computer-assisted constraint-based framework for reassembling fractured 3D objects is presented in [20]. It is addressed to cultural heritage experts, as it is an interactive system based over their experience and expertise. [21] present a puzzle engine with tools that provide 3D manipulation, reference plane alignment, segmentation, and registration. A Virtual Reality environment is also presented as matching tool, to give the user an intuitive understanding of the fragments in terms of scale, texture, materials, and other geometric attributes, thus facilitating the reassembling process.

3 DAFNE

In 2019, we threw a challenge, DAFNE (Digital Anastylosis of Frescoes challeNgE), to look for solutions that apply the most advanced techniques, such as machine learning and deep learning, to the reconstruction of destroyed frescoes. The goal was to collect the best algorithms, starting from the digitalization of the fresco broken elements, to offer restorers an additional tool for the restoration task. The faced critical issues are: i) the number of randomly mixed fragments, usually huge; ii) the corruption of the fragments, mostly with irregular shapes; iii) the mismatch of the boundaries of the collected eroded pieces; iv) the fact that some fragments have gone irretrievably lost; v) the presence of spurious/distractors elements, due to pieces of neighboring frescoes also involved in the building collapse.

Note that the competition referred to cases in which the original image of the fresco, e.g. a photo either B/W or colored, was known.

The challenge could be approached in two different ways: 1) with an automatic approach, aimed at identifying a digital process for the reconstruction of a sort of complex and incomplete puzzle; 2) with a manual (semi-assisted) approach, using a specifically developed software tool that can move the fragments images 'by hand' and get the corresponding coordinates table. As the condition of visual thinking ("think through images") is relatively widespread within the autism spectrum disorder, the second approach gave autistic people with a high cognitive ability the possibility of applying their skills in 'puzzle' solving contexts, so supporting their social involvement while fostering their inclusion in productive activities, and promoting and appreciating their potential. In fact, the tool that requires user intervention in the reassembling task, was tested and used by autistic subjects.

Figure 1. Examples of real fresco fragments.

Twenty-eight teams (grouping fifty-two people) enrolled in the competition, with participants from eight different countries (Brazil, Bulgaria, France, Germany, Italy, Portugal, United Kingdom, United States). Nine solutions were received from seventeen people, grouped into three teams of two people, a team of three, a team of four, and four individuals.

Figure 2. Top: example of plane tessellation (500 fragments, random distribution with clusters). Bottom: ground truth (original image in grey background, with overlapping colored fragments). Fragmentation applied on an image of "Domenichino, The Virgin and the Unicorn, 1602". Critical issues: lack of continuity of the collected fragments (as some pieces have gone irretrievably lost), mismatch of the boundaries of the collected pieces (due to erosion of the pieces), and interference/distractors elements (as in the same group there may be pieces of different frescoes).

The challenge entitled participants to use, at their discretion, several different cases of well-known frescoes that have been artificially 'cracked' to obtain fragments to populate the reference dataset. Starting from examples of real fragments (see Fig. 1) we based our simulated fragments generation strategy on some generic considerations: the fragments must be in different sizes, they generally do not have sharp edges, they may be convex or concave, a threshold has to be fixed in order to eliminate the fragments that result too small after the fragmentation, size and shape of the fragments must be independent from their position inside the image.

So, for the cracking process, we used five different parameters to generate the elements, and the plane tessellation depends on: i) the number of fragments, ii) the type of random distribution of fragments into the fresco plane, iii) the percentage of missing fragments, iv) the percentage of spurious fragments, and v) the ratio between the fragment area after the erosion and the original area in the fresco plane tessellation (Fig. 2).

Figure 3. Plane tessellation with a concentrated distribution adopting the Euclidean distance (image of the fresco "Entry into Jerusalem" by Giotto, 1303-1305).

Figure 4. Plane tessellation with a uniform distribution adopting the Manhattan distance (image of the fresco "Entry into Jerusalem" by Giotto, 1303-1305).

The dataset was designed to be realistic, natural and challenging for cultural heritage domains in terms of resolution, diversity in scenes, and pictorial assets. The fragments were obtained from each fresco image through a random plane tessellation with suitable statistics, with a few clusters (Fig. 3) or uniform (Fig. 4), with an erosion process applied to each fragment.

First, ANASTYLOSIS dataset DB1 was released, to be used as training set for the development phase, including for each fresco a set of fragments, eventually mixed with spurious elements (distractors) and the expected reconstructed image, so that ground truth fragment alignments were available for verification. Later, ANASTYLOSIS dataset DB2 was released for the testing phase, including, for each fresco, a set of fragments but without the corresponding solution. The whole dataset (DB1+DB2 with solutions) is now publicly available at the address https://vision.unipv.it/DAFchallenge/DAFNE_dataset. Its use is restricted to research, education and non-commercial purposes.

For each fresco an image of the original artwork and 18 folders are provided, each one containing: a set of fragments, randomly generated, eventually mixed with spurious ones; a list of the coordinates of the location and orientation of the fragments correctly placed in the reconstructed image; a list of the spurious fragments if present; the parameters used for the simulated fresco fragmentation; and the solution, consisting in the image of the original fresco with a grey background, with overlapping coloured fragments, and the corresponding coordinates table.

The winners of the challenge − assessed on the basis of the largest fresco area reconstructed correctly − presented their results and have been awarded during the 20th International Conference on Image Analysis and Processing (ICIAP 2019) held in Trento, Italy, from 9th to 13th September 2019. In accordance with the quality requirements and metrics specified in the challenge notice, two participants (one individual and one team) resulted of equal merit: a person belonging to the autism spectrum and a team of three people exploiting computer vision and pattern recognition techniques.

A new challenge, in fact an extension of the previous one, is "blind anastilosys", related to cases in which information on the frescoes content is missing, such as for example the Roman frescoes in Pompei. To this purpose, new facilities for the anastylosis package have been implemented, because it is evident that in this framework−the "blind" case −missing the original image and so missing a direct match fragment/model, make the complexity grow considerably.

4 JAVASTYLOSIS: a computer-assisted tool for assembling images of fragmented 2D frescoes

Javastylosis is a computer-assisted tool for matching images of fractured fragments, designed to assist Cultural Heritage experts and restorers in the frescoes' reconstruction process based also on the intra-fragments matches. The tool has been studied for and tested by people belonging to the autism spectrum – with high cognitive abilities and very acute visual skills, that "think through images" – as part of a project developed to give inclusion opportunities to this group of disadvantaged people. Javastylos is written in Java and the API JavaFX has been used for creating the graphical user interface (GUI). For the development, the well-known MVC (Model View Controller) software design pattern has been followed.

The GUI has been designed for achieving an easy and intuitive handling of fragments, that the user can assemble relying on chromatic and semantics-based reasoning. The user can easily move and rotate each single piece, zoom, group the fragments and operate on them as a single entity, discard pieces that are considered not belonging to the fresco to be reconstructed. Automatically a table identifying each moved digital fragment with its final location and rotation will be updated for restorers' usage during the phase of artwork reassembling with tangible fragments.

The graphical interface of Javastylos is divided in two main areas (Fig. 5). The working area consists in a big rectangular panel at the center of the screen, reserved for the recomposing attempts. A secondary panel – a horizontal bar at the bottom of the screen – shows the fragments images, that can be scrolled using the two arrows at the sides. The menu on top shows the available tools and operations on file and fragments, that the user can select from drop-down lists. The functionalities are the following: grouping fragments; toggling background color of the reference image; exporting data as image or as coordinate list (position and rotation of each fragment).

Figure 5. Javastylosis GUI. Top: no background image is available. Bottom: DAFNE solution: a transparency filter can be applied immediately to each fragment and a gray-level background image is used directly as verification.

The reassembling task starts selecting a fragments images directory and a reference picture (an image of the original undamaged fresco to be used as background for verification – if available – or a white image). Intermediate solutions can be saved in a CVS file, to be restored for following work sessions, as you can save a project and load your previous saved file.

When loading a reference picture, the user can change the view to a color version, a black and white version, or a transparency version. Then, he/she can select and move the image of a fragment – shown in the horizontal bar at the bottom of the interface – and manipulate it in the working window. Simple geometrical transformation (translation and rotation around the center) can be applied using the mouse and the keyboard shortcuts.

Additional features have been introduced to group multiple fragments into a single structure, called 'group'. Fragments belonging to a group can be translated and rotate all together. Rotation can be applied around the center of anyone of the elements that compose a group. The user can also delete groups, add and remove single fragments to a group and join groups together grading the group coordinates.

After the first basic experiments adopting this solution, derived by the DAFNE experience, it immediately resulted to us that it was necessary at least to support the user with extra facilities to find intra-fragments matches: i.e. exploitation of online features to identify and recognize chromatic and semantic matches through texture analysis.

5 Texture features

Texture can be regarded as complex visual patterns composed of entities, or sub patterns that have characteristic brightness, color, slope, roughness, size, regularity, etc. changing in space [22]. Thus, texture can be regarded as a similarity grouping in an image [23]. In general, texture is composed of primitives, with a structure consisting of repeated arrangements in space, reflecting the spatial variation of image brightness or color.

Two major issues in texture analysis have been considered: 1) to compute a characteristic of a fragment able to numerically describe its texture properties (feature extraction); 2) to determine to which of a finite number of physically defined classes a homogeneous texture region of a fragment belongs (feature classification).

The histogram of intensity levels (or RGB color) is obviously a concise and simple summary of the statistical information contained in the image. Calculation of the grey-level histogram involves single pixels. Thus, the histogram contains the first-order statistical information about the image or fragment.

Second order statistics are normally considered very useful in texture discrimination. The grey level co-occurrence matrix is one of the most basic data structure in the symbiotic texture analysis. The principle is to take a point (x; y) and another point (x + a; y + b) in the image to form a point pair ([a, b] is called the displacement). The grey value of the pair corresponds to (i, j), where i is the grey value of the point (x; y), and j the grey value of the point (x + a; y + b). Finally, the grey level co-occurrence matrix is the joint probability distributions of pairs of pixels, with a given displacement [a, b], defined as: given the image f(x,y) with a set of G discrete intensity levels, the co-occurrence matrix h[a,b] (i,j) is defined such that its (i,j)th entry is equal to the number of times that f (x1, y1) = i and f (x2, y2) = j, where x2 =x1+a e y2=y1+b.

The co-occurrence matrix contains G2 elements that is too much for texture analysis in a reasonable time. A reduced number of features can be calculated using the co-occurrence matrix for the purpose of texture discrimination [24], among the 14 defined by Haralick [25]. We are experimenting with the following four:

ASM (Energy) $A = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} h_{[a,b]}(i,j)^2$

Contrast $C = \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} (i-j)^2 h_{[a,b]}(i,j)$

Homogeneity $H = \frac{\sum_{i=0}^{G-1} \sum_{j=0}^{G-1} (ij)\, h_{[a,b]}(i,j) - \mu_x \mu_y}{\sigma_x \sigma_y}$

Entropy $E = - \sum_{i=0}^{G-1} \sum_{j=0}^{G-1} h_{[a,b]}(i,j) \ln\left(h_{[a,b]}(i,j)\right)$

In figure 6 is shown the new GUI to sort the fragments on the basis of first and second order statistical parameter.

6 Conclusions and further developments

Restoration of damaged frescoes is a crucial task in cultural heritage preservation and digital solutions can become a precious additional tool for restorers. Javastylosis is a computer-assisted system proposed for manual digital reassembling. It has been implemented to involve autistic subjects in a post-seismic recomposition of destroyed frescoes. It is a project of valuable social impact, as it can be both an inclusion opportunity as well as an enhancement of peculiarities and abilities of disadvantaged people.

Firstly, Javastylosis has been tested and evaluated, in the framework of the DAFNE project, by an autistic subject. The result was very promising. The participant was the winner of the international challenge, resulting of equal merit to a team that implemented an automatic approach. Suggestions from the user were implemented for tool's improvements, in particular related to fragments grouping facilities. In the near future the tool will be used in a second case study, for blind anastylosis applied to some frescoes from Pompei.

Pompei is a city in the province of Naples (Italy), known mainly for being buried by an eruption of Vesuvius in 79 AD. The city has been brought to light at the end of the 18th century and it is now a famous archaeological place, UNESCO World Heritage Site since 1997. Numerous artifacts and works of art have been found in Pompei, including mosaics, furnishings and other common objects of the ancient citizens. In particular, numerous frescoes – made between the end of the second century A.D. and 79 A.D. – decorate houses and commercial buildings of the city and many of them still need to be restored. The Roman frescoes are usually described as a series of four styles based on the changes in the use (or non-use) of techniques for creating perspectival depth of the wall decoration [26]. While first style decorations are imitating a wall, this wall is including a perspectival vistas in the second style, in the third style there is again a preference for closed walls, while this trend is inverted again in the fourth style, when perspectival issues return in the decoration. In fact, this classification that follows a chronological sequence can also be useful in synergic way for the content analysis and in the Pompeian frescoes anastylosis.

Figure 6. Javastylosis texture GUI. Seven features are shown: the first three belonging to the first order statistics (dimension, luminosity and yellow/blue/red component); the last four, belonging to the second order statistics, are computed over the co-occurrence metrics, and correspond to the feature's defined in the four equations of section 5 (energy, contrast, homogeneity, entropy).

While DAFNE was focused on the research of solutions having the original image of the fresco available as hint for the reconstruction process, the project TIRESIA (from the name of the mythic blind clairvoyant Theban prophet), is devoted to the reconstruction of frescoes in the absence of the previous original and complete image and with no data on frescoes content and semantics, a sort of 'blind' anastylosis. This new task may contribute to the restoration of very challenging situations, such as ancient artworks of archaeological sites.

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